

Day 175/365

**R^2 says 0.99. OOB
says 0.23.**

**The catch with random forests on tiny
data.**

Two scores, one model. Who's right?

Yesterday's forest gave two very different grades:

0.988

R^2 , tested on just **2 rows** I held back

0.23

OOB score, tested on the **training rows**, each one scored by trees that never saw it

Neither is "lying". They answer **different questions**, on different rows. The real question is which one is more trustworthy.

Why 2 test rows prove very little

Think of a 2-question quiz. Get both right and you scored 100%. Does that mean you've mastered the subject? Of course not. It depends on which 2 questions you got.

Same here. With 10 rows, an 80/20 split leaves 2 test rows. A different split could give a very different R^2 .

The test error was small, about 24,800 in salary terms. Salaries in this data climb to very high values at the top levels, so that tiny error hints the 2 test rows were from the **easy part**. You can confirm by printing `X_test`.

Why OOB was harsher

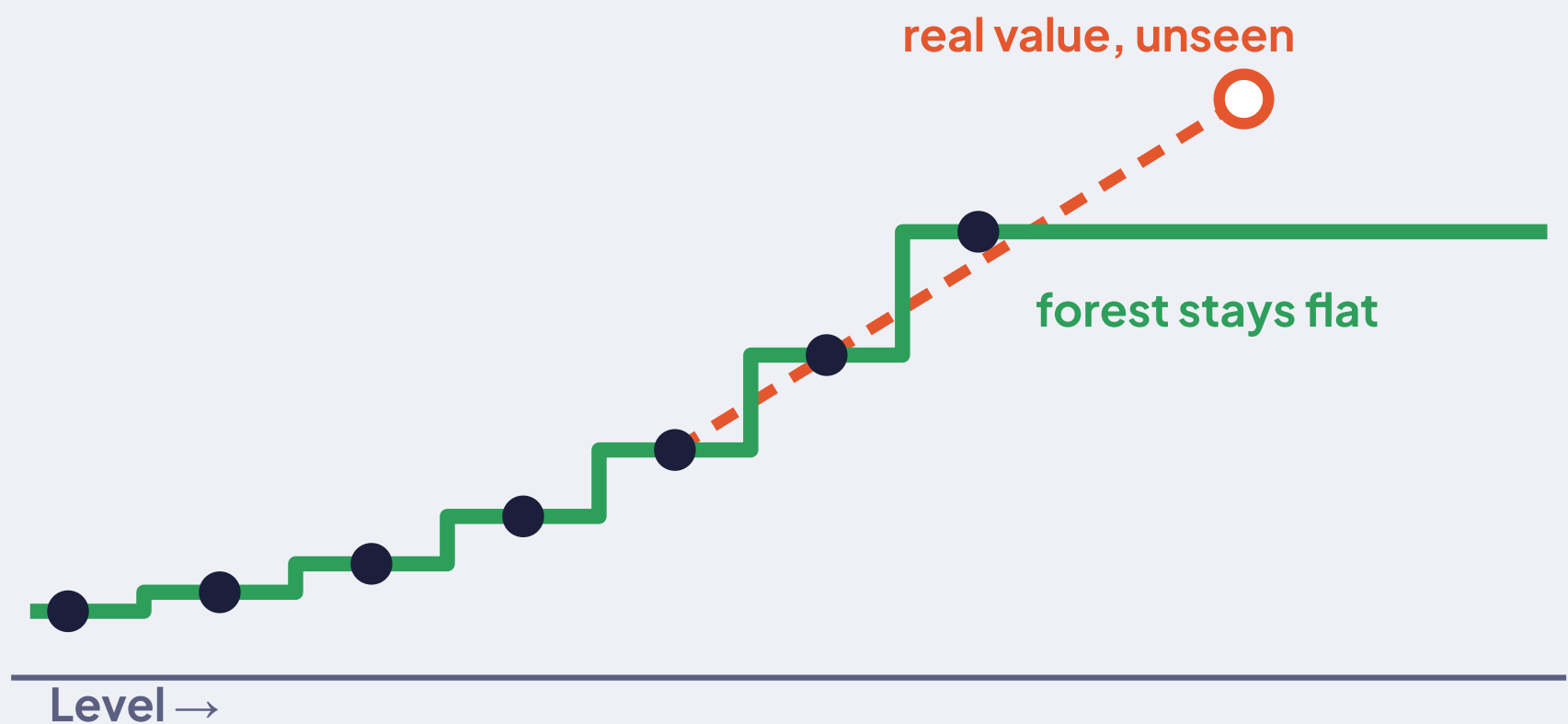
OOB checks **every training row**, including the hard ones at the top, where salary shoots far above the rest.

Hard case: a tree that didn't see the top level has nothing similar to learn from. It can only answer with salaries it has seen, which are much lower.

Easy case: the middle levels have neighbours on both sides, so every tree can do reasonably well.

So OOB, with about 8 rows each tested by the trees that skipped it, exposes the weakness that the 2-row test missed.

A forest can't go beyond what it has seen



A forest's prediction is an **average of values it saw in training**. So it can't predict higher than the highest salary it learned from.

Simplified sketch.

What I'd do instead

1 Get more data. 10 rows can't support any strong claim.

2 Use cross-validation. Test on many different splits and average, like I did with polynomial degrees on Day 167. With 10 rows, leave-one-out works.

3 Look at more than one score. R^2 , RMSE and OOB together tell a fuller story.

4 Check where it fails. The 0.23 pointed straight at the top level. A score is only useful if you follow it to the cause.

Which model, when?

	Linear / Ridge	One tree	Forest
Easy to explain	Yes	Yes	Harder
Handles curves	Needs powers	Yes	Yes
Goes beyond seen data	Yes	No	No
Needs scaling	Often	No	No
Steady when data changes	Yes	Jumpy	Steadier
Speed and memory	Light	Light	Heavier

No model wins every row. The right one depends on your data and what you need to explain.

Series recap

Random forest in 3 posts

Day 173: many different trees, averaged or voted.

Day 174: five steps in Python to predict a number.

Day 175: a great-looking score on 2 rows means little. Cross-check it, and remember a forest can't extrapolate.

When you need to explain a model to a manager, would you pick a tree, a forest or a linear model?